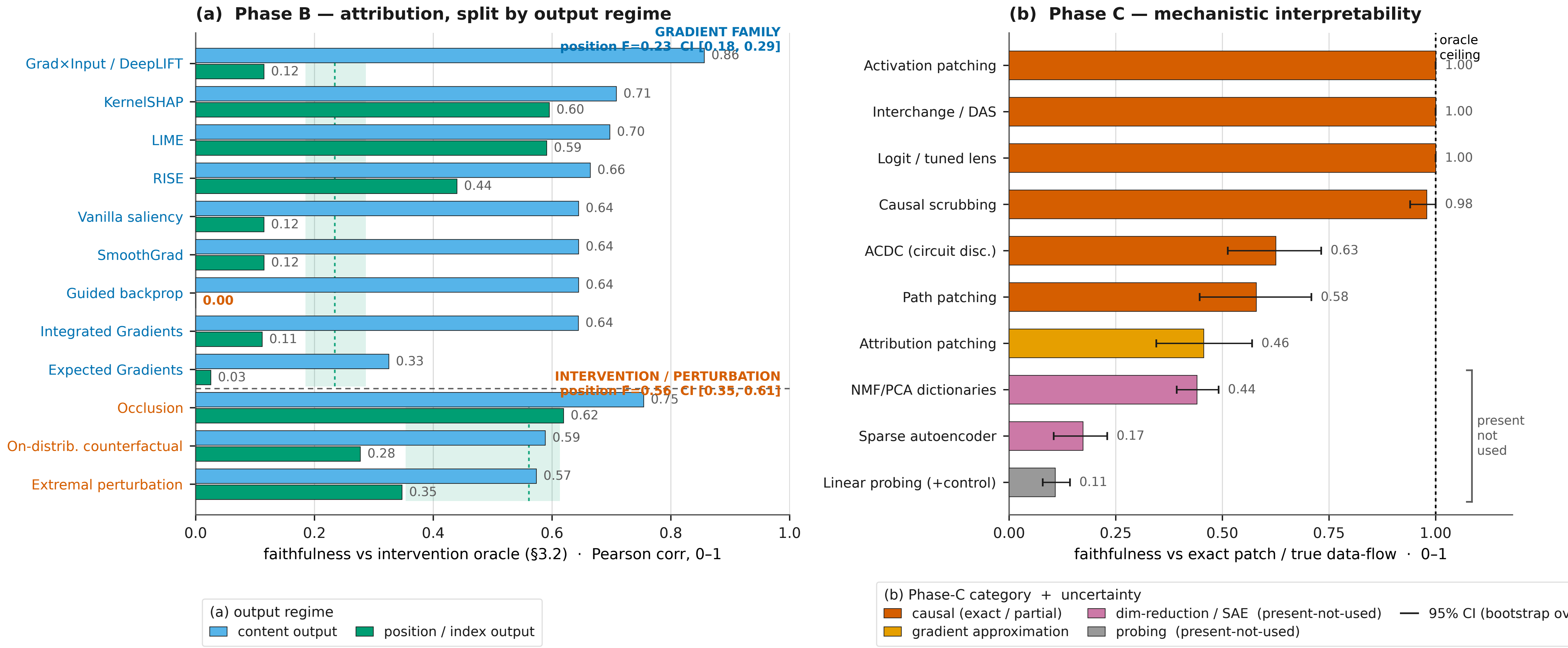
Attribution (Phase B) vs mechanistic interpretability (Phase C): faithfulness on the bit-exact VCS

The 22 Phase-B + Phase-C method rows (a subset of the 30 methods + the oracle = 31-row leaderboard; the cross-tradition scatter is Fig. 2).

HEADLINE — across ALL output regimes, intervention/causal methods are more faithful to the true mechanism than gradient/correlational ones. Causal/intervention F=0.721 (95% CI [0.695, 0.749]) vs gradient/correlational F=0.384 (95% CI [0.355, 0.412]) — a 0.337 gap, 95% CI [0.310, 0.364] (excludes 0: robust). On the POSITION/INDEX regime alone (discrete sprite-position outputs), the gap is now SIGNIFICANT on the 42-game battery: causal 0.561 vs gradient 0.234; family-mean gap 0.327, 95% CI [0.132, 0.370] over 42 games (EXCLUDES 0).



Data: committed records (leaderboard + bootstrap CIs + position bootstrap); 42-game scored battery, 30-frame horizon; pure read — no experiment re-run. Exact paths & per-number provenance in the Supplement.